

Abstract

Deep learning-based plant disease recognition systems require large and diverse image datasets to perform accurately. This project addresses the lack of diseased plant images by using an enhanced CycleGAN model to generate synthetic leaf images. Two models were tested: Model 1 (Sobel filtering on input/output) and Model 2 (threshold + Sobel on input, Sobel on output). Trained on PlantVillage healthy images, outputs were evaluated using FID, KID, and classification accuracy. Model 2 achieved better realism and boosted MobileNetV2 performance to 77.50% in a mixed real + synthetic dataset, showing the value of structure-preserving preprocessing in plant disease image synthesis.

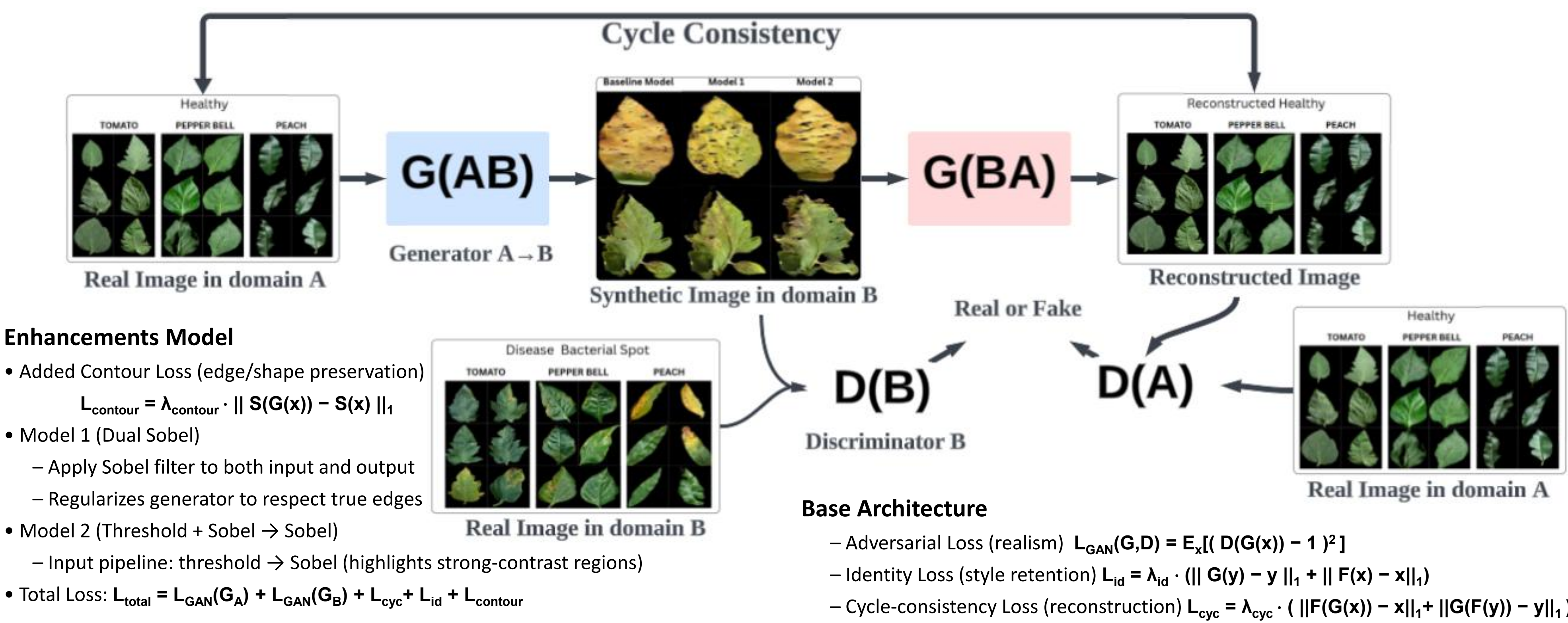
Problem Statement & Objectives

- Problem Statement**
- Lack of real diseased leaf images** - Limited dataset diversity
 - GAN models produce unrealistic results** - Fail to preserve the natural structure of leaves.
- Objectives**
- To design an image synthesis model capable of generating disease symptoms on images of healthy plant leaves.
 - To assess the quality of the synthesized images using image evaluation metrics and compare the results with existing methods.
 - The synthetic images are evaluated using application-specific metrics by introducing a new loss function to preserve leaf shape (using multiple plants with a single disease type).

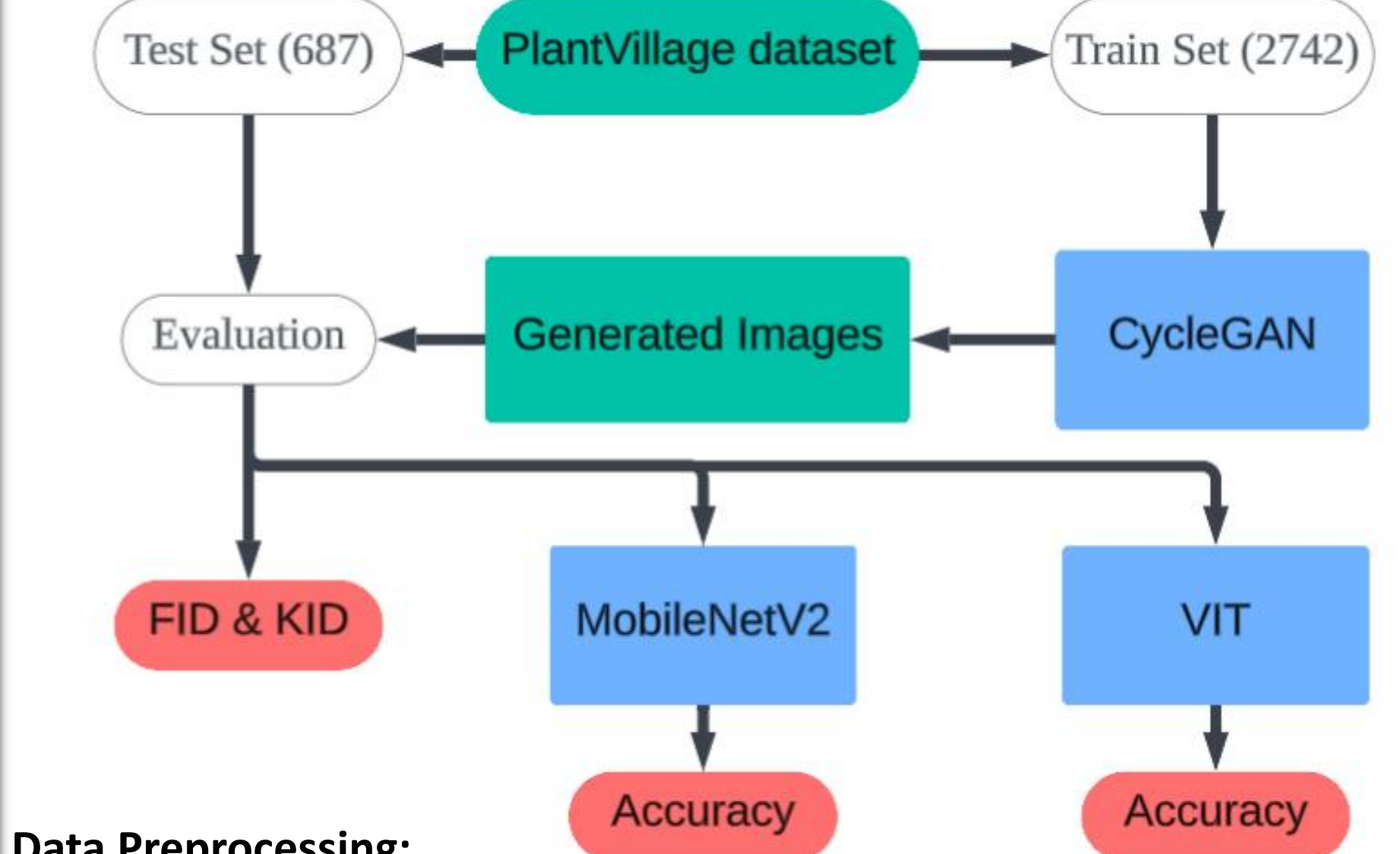
Literature Review

- PlantVillage Dataset (Mohanty et al., 2016)**
- 54000+ high-resolution leaf images covering 14 species and 38 healthy/disease classes, includes RGB, segmented, and grayscale formats.
- Original CycleGAN (Zhu et al., 2017)**
- Introduced unpaired image-to-image translation. Ideal for domains which collecting matched healthy/diseased image pairs is impractical

Implementation



Research Methodology



- Data Preprocessing:**
- PlantVillage dataset select healthy and bacterial spot of Peach, Pepper, and Tomato
 - Total trainA (2742) and trainB (4335) set is 7077.
 - Total testA (687) and testB (1086) is 1773.

- Model Evaluation:**
- Synthesized Image Quality Assessment: KID & FID score
- $$FID = ||\mu_r - \mu_g||^2 + T_r \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2} \right) \quad k(x, y) = \left(\frac{1}{d} x^T y + 1 \right)^3$$
- Binary classification (Healthy vs. Diseased):**
- Classifier used: MobileNetV2 & Vision Transformer (Accuracy)
 - All Real: 2742 real images per class
 - Real + Synthetic: 1371 real + 1371 synthetic images per class
 - All Synthetic: 2742 synthetic images per class

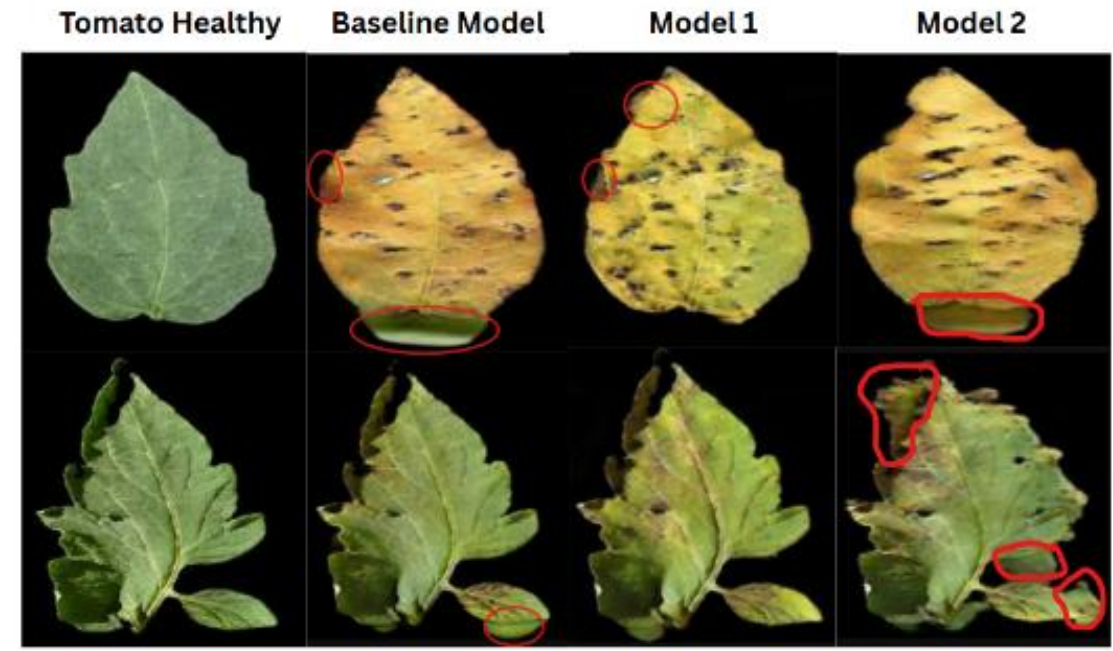
Evaluation

Image Quality Evaluation

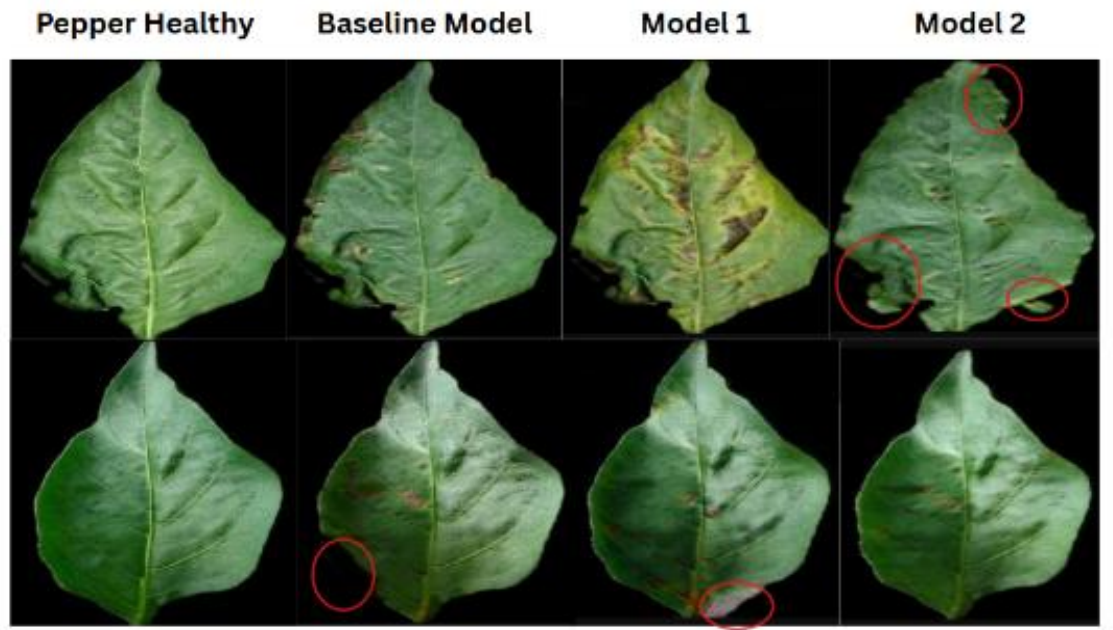
CycleGAN Model	FID	KID
Baseline Model	33.32	0.67
Model 1 (Dual Sobel)	41.16	0.67
Model 2 (Threshold + Sobel → Sobel)	37.95	0.68

Classification Accuracy on Synthetic-Augmented Datasets

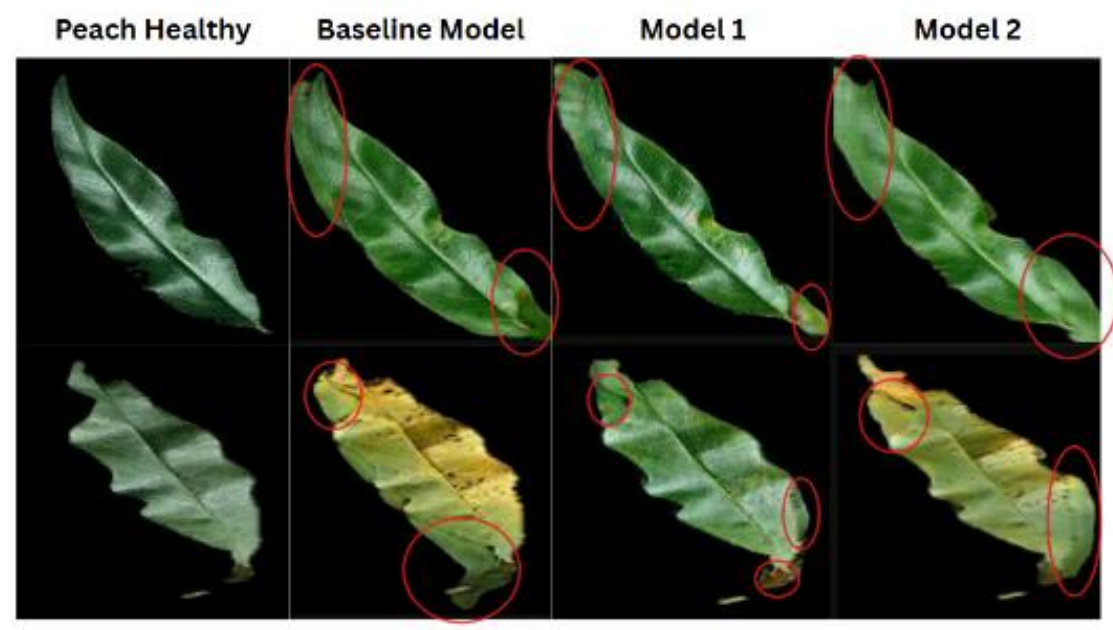
Dataset Type	VIT Accuracy	MobileNetV2 Accuracy
All Real	98.08	72.25
Real + Synthetic (Model 1)	96.28	41.62
Real + Synthetic (Model 2)	95.09	77.50
All Synthetic (Model 1)	93.91	38.86
All Synthetic (Model 2)	95.54	61.25



- <- Tomato**
- Model 1: Strong shape fidelity but overly dense lesion clusters.
 - Model 2: More natural textures, but some lesions appear artificial (striped or pit-like).
 - Baseline: Realistic scattered necrosis, good vein structure, but blur at edges.
 - Summary: Baseline gives best lesion realism; Model 1 is too dense; Model 2 trades shape for texture.



- <- Bell Pepper**
- Model 1: Best shape preservation, vivid but exaggerated disease (burn-like).
 - Model 2: Natural symptom texture (top), but weak and inconsistent symptoms (bottom), worst shape distortion.
 - Baseline: Consistent shape with mild distortions, repetitive synthetic-looking spots.
 - Summary: Model 1 shows clear disease, Model 2 has better realism (top row only), but at the cost of leaf shape.



- <- Peach**
- Model 1: Sharp vein detail, realistic plaques (top), but introduces unnatural red patches (bottom).
 - Model 2: Better yellowing (bottom), smoother shapes than Model 1, but adds phantom leaflets and lacks clear plaque texture.
 - Baseline: Balanced shapes and smooth green tone but lacks strong disease realism.
 - Baseline is moderate; Model 1 exaggerates; Model 2 better color but distorts structure.

Conclusion

- Two Sobel-enhanced CycleGAN variants were compared:
- Model 1 (Sobel→Sobel) preserves edges but over-smooths disease patterns, yielding higher FID (41.16).
 - Model 2 (Thresh + Sobel→Sobel) balances contrast and structure preservation, cutting FID to 37.95 and boosting MobileNetV2 accuracy to 77.5%.
- Synthetic augmentation from Model 2 significantly improves lightweight classifiers in mixed real+synthetic settings, though heavyweight ViT still excels on real-only data.
- Structure-preserving preprocessing (threshold + Sobel) proves key to realistic symptom rendering without shape distortion.